

A comparison of different types of cloud data management storage

Storage type	Description	Use cases	Examples
Object storage	Stores data as objects rather than files or blocks.	Storing large amounts of unstructured data like photos, videos, and backup files.	Amazon S3, Google Cloud Storage, Azure Blob Storage
File storage	Provides a traditional file-based interface. Suitable for shared file systems and NAS.	Applications requiring a file system interface, such as network attached storage (NAS) or file servers.	Amazon EFS, Google Cloud Filestore, Azure Files
Block storage	Stores data in volumes or blocks, each with its own address. Suitable for high-performance data.	Storing data requiring high performance and low latency, such as databases and virtual machine disks.	Amazon EBS, Google Cloud Persistent Disk, Azure Disk Storage
Database storage	Specifically designed for storing structured data in a DBMS. Optimised for fast and efficient access.	Applications requiring frequent access to structured data.	Amazon RDS, Google Cloud SQL, Azure SQL Database
Cold storage	Designed for storing infrequently accessed data for long periods. Cost-effective but with higher latency.	Storing data that needs to be retained but is rarely accessed, like archives.	Amazon Glacier, Google Cloud Storage Coldline, Azure Archive Storage
Hybrid storage	Combines on-premises storage with cloud storage for scalability and flexibility.	Organisations wanting cloud benefits while maintaining control over their data.	NetApp, Dell EMC

on legal and business requirements.

- **Compliance:** Ensure your data management practices adhere to relevant regulations like GDPR or HIPAA.

Data architecture

A well-designed data architecture is the foundation for scalability and flexibility in the cloud. Here's what to consider.

- **Scalability:** Choose an architecture that can easily scale up or down as your data volumes fluctuate.
- **Flexibility:** Design your architecture to accommodate new data sources and evolving business needs.
- **Cloud service model:** Select the appropriate cloud service model (IaaS, PaaS, SaaS) based on your desired level of control and infrastructure management.

Data integration

Seamless data integration across various cloud platforms and on-premises

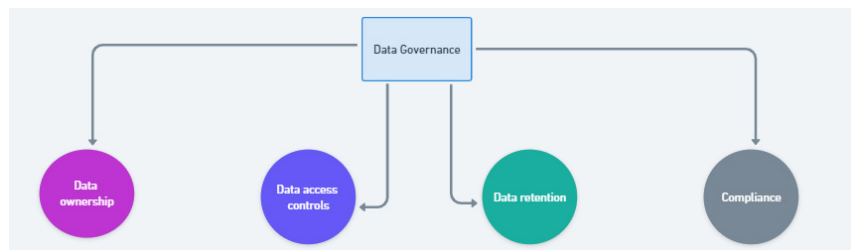


Figure 2: Data governance

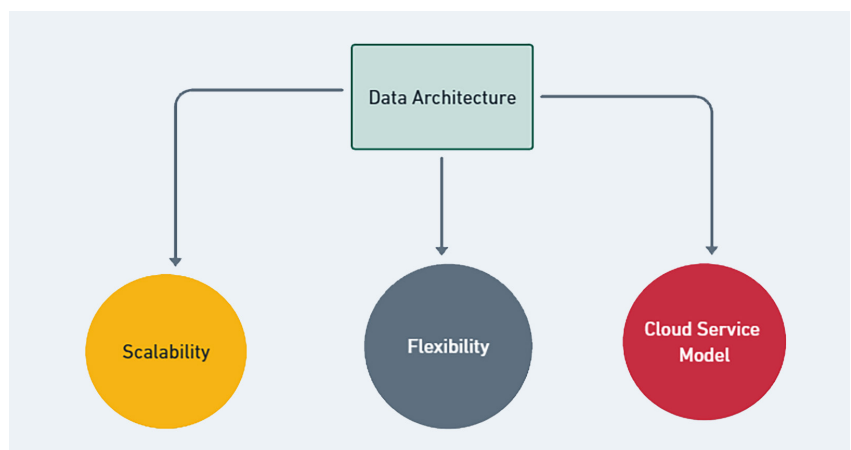


Figure 3: Data architecture